PawMatch Pretesting Development Test Plan

OCR A Level Computer Science H446 Programming Project

Project	PawMatch animal adoption matching system
System	Python, Streamlit and SQLite
Purpose	Planned test data for iterative development, with a separate post-development plan
Status	Pretesting plan. Actual results, evidence, failures, fixes and retests must be completed while development takes place.

Before implementing the solution, I identified test data for each planned iteration so that every module can be checked as it is developed. The tests are linked to the five decomposed requirements in my design: R1 authentication, R2 the adopter homepage, R3 administrator animal management, R4 animal browsing and R5 the questionnaire and recommendations. I chose normal, boundary, erroneous, security, persistence and algorithm-oracle data because each category exposes a different type of defect.

This plan supports the OCR design requirement to identify and justify iterative and post-development test data. It does not by itself prove that testing occurred. During development I must add dated, annotated evidence at every stage, including failed tests, the cause, the remedial action, why that action was chosen and the result of the retest. Final robustness and stakeholder usability testing must remain separate from development testing.

This document is structured to support top-band evidence, but no document can guarantee a particular mark. Marks depend on the quality and authenticity of the completed development record and the final submitted project.

OCR assessment alignment

Assessment evidence	Top-band expectation	How this plan supports it
AO3.1 Design	Identify and justify test data used during iterative development and further data for post-development testing.	Every case supplies concrete data, an expected result and a reason for choosing that class; post-development cases are listed separately.
AO3.2 Development	Evidence of testing at every iteration; failed tests and remedial actions with full justification.	Seven implementation stages and evidence prompts provide a structure to complete contemporaneously. Blank results must be replaced with authentic evidence.
AO3.3 Evaluation	Separate annotated functional and robustness tests, plus usability testing, then evaluation against success criteria.	A reserved final-testing section prevents development tests from being reused as evaluation evidence and includes adopter/admin feedback.

Testing method and evidence rules

Test class	Why it is included	How I will apply it
Normal	Shows that realistic data produces the intended outcome.	Use representative adopter, animal and questionnaire records.
Boundary	Defects often occur at the exact accepted limit or immediately outside it.	Test lower/upper limits and values just below/above them.
Erroneous	Shows invalid, missing or unsupported values are rejected safely.	Use blank, malformed, wrong-type and unsupported lookup data.
Algorithm oracle	The program output needs an independently calculated expected value.	Show the weighted points and percentage calculation before running the code.
Security and integrity	Authentication, roles, parameterised SQL and foreign keys protect data.	Use synthetic accounts, direct URL attempts, SQL-like text and foreign-key checks.
Persistence and state	Streamlit reruns and restarts must not lose or mix records.	Compare visible state with SQLite rows before/after reruns and restarts.
Regression	A fix can break previously working features.	Repeat the failed test and a small related critical pack after each change.

Evidence rules

1. Give each screenshot, console log or short video frame the same test ID as the plan.
2. Record the build/iteration and date when the test was actually run. Do not recreate the evidence at the end.
3. Show the input, visible or logged output and the expected-versus-actual comparison. A results table without evidence is not enough.
4. Preserve failed tests. Explain the cause, the chosen fix and why it was preferable, then show a passed retest and related regression tests.
5. Use only synthetic personal data. Redact password hashes, local paths or anything that identifies a real user.
6. Keep final evaluation tests distinct. Development evidence shows how the build changed; evaluation evidence judges the finished solution.

Planned iterative development tests

The order below follows the dependency of the actual PawMatch system. Each stage can be implemented, tested and reviewed before the next stage depends on it.

Iteration 1 Authentication and user database

R1 User and administrator authentication. Build the users table, registration validation, password hashing, login and role-based redirection before any protected pages depend on them.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
A01 Q9·Q12	R1 Registration	Normal functional data. Proves a realistic adopter can create an account.	Register name 'Test Adopter', email 'new.user@example.com', age 24, password 'abcdefghi', matching confirmation.	One user row is created with role 'user'; a success message is shown; the user can proceed to login.	Registration screen and database row with password value hidden/redacted.
A02 Q9·Q15	R1 Name validation	Boundary and erroneous data. The documented valid length is 2 to 50 characters.	Try 1, 2, 50 and 51 character names; blank and whitespace-only names; a name containing '@'.	Lengths 2 and 50 are accepted. The other values are rejected with a specific message and no row is inserted.	Each boundary message plus a query showing the users table is unchanged after rejection.
A03 Q9·Q12·Q15	R1 Email validation	Normal, erroneous and duplicate data. Email is the unique adopter identifier.	Try 'person@example.com', 'person@', blank, mixed-case duplicate 'PERSON@EXAMPLE.COM', and surrounding spaces.	Valid email is normalised to lower case; malformed/blank/duplicate values are rejected; spaces are trimmed; no duplicate row exists.	Form feedback and a case-insensitive database lookup.
A04 Q9·Q15	R1 Age validation	Lower and upper boundary data protects the 18 to 120 database constraint.	Use ages 17, 18, 120, 121 and no value.	18 and 120 are accepted. 17, 121 and missing input are rejected before insertion.	Messages for each rejected value and accepted boundary records in a disposable database.
A05 Q9·Q12·Q15	R1 Password validation	Boundary and erroneous data. Confirms more than 8 and no more than 128 characters, plus confirmation.	Use lengths 8, 9, 128 and 129; blank password; mismatched confirmation.	Lengths 9 and 128 with matching confirmation are accepted. All other cases show the correct message and create no account.	Validation output for each boundary and a no-insert query for failures.
A06 Q12	R1 Password security	Security test. Plain passwords must never be stored and salts should prevent equal hashes.	Create two disposable users with the same password. Inspect password_hash values without exposing the password in screenshots.	Both stored values use the scrypt format, differ from the password and differ from each other; both verify correctly.	Redacted database output showing distinct hashes and verification assertions.
A07 Q12	R1 Login	Normal and erroneous data verifies authentication without revealing which credential failed.	Login with valid mixed-case email and correct password; then valid email/wrong password and unknown email/correct-looking password.	Valid login creates a user session and opens the home page. Both invalid attempts show the same generic error and remain on login.	Login states and session keys; error wording for both failures.
A08 Q5·Q12	R1 Admin access	Security and role test prevents privilege escalation.	Login with the seeded admin username; login as a normal user and navigate directly to the admin page URL.	Admin reaches the dashboard. A normal or logged-out user is denied and redirected; the user cannot add, edit or delete records.	Role/session values and both redirect outcomes.

Iteration 2 Session handling and adopter navigation

R2 User homepage and main menu. Add protected navigation only after authentication is reliable.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
N01 Q12	R2 Protected pages	Erroneous state test. Direct URLs must not bypass login.	With a cleared session, open Home, Questionnaire, Browse Animals, My Account and My Matches directly.	Every protected page returns to Login without showing private data.	One screenshot per redirect or a short annotated sequence.
N02 E1·E2 E3·S6	R2 Main navigation	Normal integration test proves each menu route is connected.	Login as an adopter and select Home, Questionnaire, Browse Animals, My Account and My Matches.	Each control opens the intended page and keeps the same user session.	Navigation sequence with page headings and user ID/session continuity.
N03 Q7·S1·S2	R2 Personalisation	Normal database retrieval test links the correct user to displayed information.	Create two users with different names; login as each in turn.	The home/account pages show only the current user's name, email and questionnaire summary.	Side-by-side annotated screenshots using synthetic accounts.
N04 Q12	R2 Logout	Security regression test prevents session reuse on a shared device.	Login, choose Logout, confirm it, then use browser Back and direct protected URLs.	Session keys are cleared and protected content cannot be recovered without signing in again.	Before/after session state and redirect after Back/direct URL.
N05 Q12·Q15	R2 Rerun state	Streamlit-specific robustness test checks that expected state survives reruns without leaking between users.	Change questionnaire progress, trigger a normal Streamlit rerun, logout and login as another user.	The first user's permitted state persists across their rerun; user-specific state is cleared or replaced after logout.	Session-state values before/after rerun and after account switch.

Iteration 3 Administrator animal records

R3 Administrator menu and the animals table. Test validation and each database operation before the matching algorithm relies on these records.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
D01 Q2·Q5·Q15	R3 Add animal	Normal CRUD test supplies a complete valid record.	Add 'Nova', Dog, breed 'Labrador cross', age 2.5, Female, Medium, Moderate activity, valid care fields, 4 hours alone, Available, 20+ character description.	A single row is committed, confirmation appears, and the same values are visible after a rerun/restart.	Completed form, success message and retrieved row.
D02 Q2·Q15	R3 Animal name	Boundary and erroneous data covers 1 to 50 characters and the permitted character set.	Use lengths 0, 1, 50 and 51; valid 'K9-Rose'; invalid 'Rose@'.	Lengths 1 and 50 and permitted letters/numbers/spaces/hyphen/apostrophe are accepted. Blank, 51 and '@' are rejected.	Validation messages and no-insert evidence for failures.
D03 Q2·Q15	R3 Breed	Boundary and erroneous data covers 2 to 50 characters and prevents symbols-only content.	Use lengths 1, 2, 50 and 51; '--'; 'Labrador 2nd cross'.	2 and 50 with alphanumeric content are accepted. Other cases are rejected with no database change.	Messages and cleaned breed value.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
D04 Q2·Q15	R3 Age	Boundary and extension test preserves the existing half-year interface while enforcing 0 to 40.	Use -0.5, 0, 1.5, 40 and 40.5.	0, 1.5 and 40 are accepted. Values below 0 or above 40 are rejected.	Form messages and stored REAL values for accepted cases.
D05 Q10·Q15	R3 Hours alone	Boundary and type test covers the 0 to 24 whole-number constraint.	Use -1, 0, 24, 25 and non-numeric input at the backend validator.	0 and 24 are accepted. Other values are rejected; invalid data is not committed.	Validator output and database constraint outcome.
D06 Q2·Q15·S4	R3 Description	Boundary test covers the designed 20 to 1000 character personality description.	Use lengths 19, 20, 1000 and 1001.	20 and 1000 are accepted; 19 and 1001 are rejected with clear messages.	Character counts and messages.
D07 Q2·Q15	R3 Lookup fields	Erroneous backend test ensures UI controls are not the only validation.	Call validation with unsupported species, sex, size, activity, home type, child suitability, experience and status values; pass invalid booleans/ranges.	Every unsupported value is reported and no record is inserted.	Error dictionary and unchanged row count.
D08 Q5·Q15	R3 Edit	Normal persistence test verifies update by unique animal ID.	Edit Nova's activity, description and status; keep animal_id unchanged; restart or rerun.	Only the selected row changes, updated_at changes, and the edited values persist.	Before/after query and refreshed dashboard.
D09 Q5·Q12·Q15	R3 Delete	Normal, cancellation and referential-integrity test covers a permanent operation.	Choose Delete then cancel; repeat and confirm for a disposable animal that has a match_result.	Cancel leaves all data unchanged. Confirm removes the animal and dependent match row without orphan foreign keys.	Row counts, confirmation control and PRAGMA foreign_key_check output.
D10 Q12·Q15	R3 SQL safety	Security/erroneous data test checks parameterised statements.	Search for a string containing a quote and SQL-like text such as "' OR 1=1 --". Submit it only through fields that permit the characters.	Text is treated as data; no unexpected rows are returned/changed and all tables remain present.	Search result, table list and row counts before/after.
D11 Q5·Q15	R3 Schema migration	Regression test protects existing project data when the old 12-hour constraint is upgraded.	Copy a database whose animals table uses 0 to 12 hours, initialise the current schema, then inspect rows and foreign keys.	Constraint becomes 0 to 24; all animal rows are preserved; foreign_key_check returns no violations.	Schema SQL, before/after counts and migration test output.

Iteration 4 Animal browsing and profiles

R4 Animal viewing. Verify search, filters, empty states and ID-based profile selection against controlled records.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
B01 E1·E2·E3	R4 Empty search	Normal equivalence case. An empty search is designed to show all eligible records.	Leave search blank with All species and default filters.	All records allowed by the page's status rule are shown; no validation error appears.	Visible card count compared with a database count.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
B02 Q8·S6	R4 Name search	Normal and case-insensitive data verifies text normalisation.	Search ' luna ' and 'LUNA'.	Spaces are trimmed and both inputs return Luna rather than no results.	Entered value, cleaned value and resulting card.
B03 E1·E2 E3·Q8	R4 Breed search	Normal partial-match data checks the second searchable field.	Search a distinctive fragment from a seeded breed, with different letter case.	Only animals whose name or breed contains the fragment are returned.	Query/input and matching cards.
B04 Q8·Q15	R4 Search length	Boundary data covers the documented maximum of 20 characters.	Submit a 20-character search and a 21-character search through the validator.	20 characters are accepted. 21 produces 'Search cannot exceed 20 characters' and no query is run with it.	Boundary inputs and validator result.
B05 E2·Q8	R4 Species filter	Normal and erroneous lookup data checks filtering and backend defence.	Select Dog, Cat, Rabbit and All; call validator with 'Horse'.	Each valid choice returns only the intended set; All removes the condition; Horse is rejected.	Cards/species values and invalid lookup error.
B06 E2·Q8·S6	R4 Combined query	Integration test exercises search plus species together.	Use a valid name/breed fragment with a matching species, then a non-matching species.	Matching combination returns expected records. Non-matching combination shows the designed empty state without crashing.	Expected IDs from SQL and displayed IDs/empty message.
B07 E1·E2 E3·Q2	R4 Profile selection	Identity test ensures duplicate/similar names cannot open the wrong record.	Create two synthetic animals with similar names but different IDs; open the second card.	The profile is loaded by selected animal_id and shows only the second record's details.	Selected ID/query parameter and profile fields.
B08 Q6·Q8·S6	R4 Clear filters	State test verifies the reset control.	Apply a search/species filter, then select Clear filters.	Search becomes blank, species returns to All and the full permitted list reappears.	Before/after widget state and result count.

Iteration 5 Questionnaire validation and storage

R5 Questionnaire and recommendations. Validate each answer and its relationship to the logged-in user before calculating matches.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
Q01 Q1·S3·S5	R5 Valid response	Normal functional test uses one permitted value for every question.	Housing 2, garden Yes, hours 4, children No, activity 2, species Cat, size Medium.	No errors are returned and one questionnaire_responses row is stored for the current user.	Completed review screen and inserted row.
Q02 Q1·Q10·S5	R5 Level fields	Boundary and erroneous data covers housing/activity values 1 to 3.	For both fields use 0, 1, 3, 4, missing and non-numeric backend values.	1 and 3 are accepted; all other values are rejected with field-specific messages.	Validator results for both fields.
Q03 Q1·Q10·S5	R5 Hours alone	Boundary and type data covers 0 to 24 whole hours.	Use -1, 0, 24, 25, missing and 'two'.	0 and 24 are accepted. All other data is rejected and cannot be stored.	Error output and no-insert evidence.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
Q04 Q1·Q15·S5	R5 Boolean answers	Normal and erroneous backend data checks garden and children values.	For each field use True, False, 1, 0, None and 'Yes'.	True/False/1/0 are normalised to booleans. None and string data are rejected.	Cleaned values and error dictionary.
Q05 E3·Q1·S3	R5 Preferences	Normal and erroneous lookup data covers all supported choices and No preference.	Try Dog, Cat, Rabbit and No preference; Small, Medium, Large and No preference; unsupported values.	Every listed option is accepted. Unsupported values are rejected before matching.	Validator output for each equivalence group.
Q06 Q12·Q15	R5 User relationship	Database integrity test prevents one adopter's answers appearing under another account.	Create two users and save different responses while logged in as each.	Each response foreign key points to its own user and each account retrieves only its latest response.	Joined query using synthetic accounts.
Q07 Q15	R5 Latest response	Sequence/regression test checks repeat questionnaires.	Submit two valid responses for one user with different values and timestamps/IDs.	Both rows remain available for history, but the homepage and match page use the newest response.	Ordered response query and displayed summary.
Q08 E3·Q15 S6·S8	R5 Incomplete journey	Erroneous usability test ensures a user cannot submit before all seven answers exist.	Attempt to advance/submit with one required answer missing in each section.	The relevant section remains active or shows a clear prompt; no response or match rows are inserted.	Prompt and row counts.

Iteration 6 Compatibility and ranking algorithm

R5 Weighted matching. Use controlled adopter and animal dictionaries so the expected score is calculated independently before comparing it with program output.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
M01 Q10·Q15 S5·S7	R5 Maximum score	Algorithm oracle. All five welfare requirements are met.	Controlled profile meets hours 20, children 15, housing 15, activity 12 and garden 10, total 72.	Compatibility is round(72/72 x 100, 1) = 100.0, category Strong, with five matched explanations.	Hand calculation beside returned result.
M02 Q3·Q10 S5·S9	R5 Below cutoff	Boundary-near oracle. The closest planned score below the 70% threshold is 50 points.	Meet hours, children and housing only: 20 + 15 + 15 = 50 points.	Compatibility is 69.4%; preference score is 0; the animal is not stored or displayed as a recommendation.	Hand calculation, calculated result and absence from match_results.
M03 Q10·S5·S7	R5 Above cutoff	Boundary-near oracle. A 52-point combination is just above the 70% threshold.	Fail hours; meet children, housing, activity and garden: 15 + 15 + 12 + 10 = 52.	Compatibility is 72.2%; the animal is eligible and stored.	Hand calculation and stored row.
M04 Q10·S5	R5 Individual weights	Structural unit test isolates every selection branch.	Change one condition at a time so each of 20, 15, 15, 12 and 10 points is independently gained/lost.	Each change alters total_score only by its documented weight and moves the correct explanation between met/unmet lists.	Parameterised assertion output for all five factors.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
M05 Q10·S3·S5	R5 Preference bonus	Algorithm separation test ensures preferences rank but do not inflate welfare compatibility.	For the same eligible animal calculate with matching species/size, then No preference.	Compatibility is identical. Exact species adds 10, exact size adds 5, both add 15 to preference_score only.	Two result dictionaries with sensitive data removed.
M06 Q10·S3	R5 No preference	Normal equivalence test verifies neutral preferences.	Use No preference for species and size against several eligible animals.	No animal receives a preference bonus; order is decided by compatibility then deterministic ID tie-break.	Rank keys in displayed order.
M07 Q10·Q14·S7	R5 Ranking order	Algorithm oracle confirms the documented primary and secondary sort keys.	Create eligible A with preference 15 and compatibility 72.2; eligible B with preference 0 and compatibility 100.0.	A ranks before B because preference_score is the first key, followed by compatibility_score.	Input/result table and ordered IDs.
M08 Q5·Q15·S9	R5 Availability	Normal and erroneous-status test prevents unavailable animals being recommended.	Create otherwise identical Available, Reserved and Adopted animals.	Only the Available animal is assessed/stored for a new response.	Status query and match_results rows.
M09 Q14·S7	R5 Top five	Boundary display test covers fewer than, exactly and more than five eligible results.	Prepare 0, 3, 5 and 6 eligible animals.	Empty guidance appears for 0; all are shown for 3/5; only the first five ranked results are shown for 6.	Counts and ordered names for each dataset.
M10 E3·Q3 Q14·S7	R5 Explanations	Usability/correctness test makes the score auditable to the adopter.	Use an answer set that meets some requirements and fails hours, children or garden.	Matched and unmet explanations state the user's value and the animal requirement and agree with the awarded points.	Score breakdown and rendered explanation text.
M11 Q10·Q15	R5 Storage/order	Database integration test checks persistence and uniqueness.	Calculate the same response once, inspect rows/order, then recalculate it.	One row per response_id/animal_id remains; scores are 0 to 100; retrieval preserves preference-first order without duplicates.	UNIQUE constraint/query output and ordered retrieval.

Iteration 7 Integration and regression

R1 to R5 complete system. Re-run critical earlier cases after modules are connected, while retaining failed-test evidence from each earlier iteration.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
I01 E1·E2·E3 Q14·S6·S7	R1-R5 Adopter journey	End-to-end normal test proves the decomposed modules form a complete solution.	Register a synthetic adopter, login, browse, submit questionnaire, view ranked matches, open a profile, view account and logout.	Every transition succeeds; correct user/animal data persists; only eligible matches appear; logout protects pages.	Annotated sequence with database IDs and no real personal data.
I02 Q2·Q5 Q10·Q15	R3-R5 Admin to matcher	End-to-end data-flow test connects admin CRUD to adopter recommendations.	Admin adds an Available animal with known requirements; adopter submits answers predicted to score 100; admin later marks it Reserved.	Animal appears and scores 100 while Available; it is absent from newly calculated recommendations after Reserved.	Admin form, hand calculation, match page and status change.

ID	Requirement	Test type and justification	Planned data and method	Expected result	Evidence to capture
I03 Q12·Q15	R1-R5 Persistence	Restart robustness test checks that SQLite, not temporary UI state, is the source of truth.	Create user, animal, response and results; fully stop/restart Streamlit; login again.	Records and newest recommendations remain available and linked to the correct account.	Before/after restart queries and page views.
I04 Q12·Q15	R1-R5 User isolation	Security/integration test prevents cross-account disclosure.	Two adopters submit different answers; alternate sessions and direct URLs/query parameters.	Each sees only their own account, latest response and results; admin-only functions remain unavailable.	Joined database query and both account views.
I05 Q5·Q15	R3 Setup repeatability	Regression test prevents duplicate demo data during repeated setup.	Run the setup/seed routine twice on a clean disposable database.	First run creates the intended records; second run adds no duplicate admin or animals.	Console output and row counts after each run.
I06 Q15	R1-R5 Regression pack	Risk-based regression rechecks neighbouring features after every remedial change.	Run the automated authentication/database/matching suite plus A07, D08, B06, Q06 and M07 after a fix.	All previously passing critical tests remain passing; any new failure becomes a documented test/fix/retest cycle.	Timestamped test-run output and links to affected evidence IDs.
I07 Q12·Q15	R1-R5 Failure handling	Robustness test checks controlled behaviour when the database is temporarily unavailable.	Use a disposable invalid/unavailable database path or controlled mock; attempt login, registration and admin save.	The app reports an understandable error, does not expose a traceback to the user and does not claim data was saved.	User-facing message and redacted console exception.

Reusable development evidence record

Copy this record beneath each implemented feature or use the same headings in the development journal. Complete it immediately after the test. For a failure, retain the first result rather than replacing it with only the later pass.

Field	Entry to complete during development
Test ID and build	Example format: M03, Iteration 6 prototype 2, date/time
Reason for this test now	Link it to the feature just implemented and explain the risk it checks.
Input/setup used	Record exact values, account/animal IDs and relevant state. Use synthetic data.
Expected result	Copy the planned result or explain an approved change to the expectation.
Actual result	Describe exactly what happened; include calculated/database values rather than only 'worked'.
Evidence reference	Screenshot/video/log filename and annotation explaining what it proves.
Outcome	Pass or Fail, with expected-versus-actual comparison.
Cause of failure	Identify the code, query, state or design cause. Leave blank only for an initial pass.
Remedial action and justification	State what changed, alternatives considered and why this fix was chosen.
Retest	Retest ID, new actual result, evidence reference and outcome.
Regression review	Related earlier tests rerun and their results; note any new issue for the next iteration.

Minimum evidence bundle for each iteration

- A labelled prototype or code/interface screenshot showing what existed at that stage.
- Completed test records for the cases run in that stage, including expected and actual outcomes.
- At least the genuine failures encountered, not manufactured failures, with cause, fix, justification and retest.
- A short review explaining what the results changed or confirmed before the next iteration.

Separate post-development test data plan

These cases are reserved for the finished product and should not replace the development tests above. They use fresh data where possible and provide the functional, robustness and usability evidence needed for evaluation against the success criteria.

ID	Area	Further data and justification	Expected evidence/outcome
P01 Q14·Q15 S7·S8	Functional acceptance	Use a fresh synthetic adopter and animal dataset not used during development.	Complete the full adopter and admin journeys and compare every outcome with R1-R5.
P02 Q12·Q15·S9	Robustness boundaries	Use a fresh matrix of valid boundaries and just-outside invalid values.	No crash, invalid database write or misleading success message; helpful feedback appears.
P03 Q15	Stress and volume	Load a copy of the database with a substantially larger synthetic animal set and repeat browse/match actions.	Record response time, result correctness and whether the interface remains usable; judge against the stated success criterion.
P04 Q6·Q13·S6	Browser/device	Run the finished app in the browsers/devices available to the stakeholder, including a narrow viewport.	Core pages remain readable and controls usable; document any layout limitation.
P05 Q6·Q14·S1 S2·S6	Adopter usability	Ask an intended adopter to complete registration, browsing, questionnaire and match interpretation without coaching.	Record task completion, errors, comments and whether scores/explanations are understood.
P06 Q5·Q6·Q12	Administrator usability	Ask a shelter/admin stakeholder to add, locate, edit and safely delete a disposable record.	Record task completion, errors and feedback on labels, confirmation and data fields.
P07 Q6·S6	Accessibility review	Keyboard-only navigation, zoom/text scaling and contrast/readability check on the finished interface.	Record which key tasks are accessible and any barriers requiring future development.
P08 Q3·Q14·Q15 S7·S8·S9	Success-criteria evaluation	Use new post-development evidence plus relevant development evidence, without counting one test twice.	For each success criterion conclude fully met, partially met or not met and justify using evidence IDs.

Development traceability summary

Requirement	Planned development evidence
R1 User/admin authentication	A01-A08, N01, N04, I01, I04 and I07
R2 User homepage/main menu	N01-N05, I01, I03 and I04
R3 Administrator menu	D01-D11, I02, I03, I05 and I07
R4 Animal viewing	B01-B08, I01, I02 and I03
R5 Questionnaire/recommendations	Q01-Q08, M01-M11 and I01-I04

Sources used to shape the plan

OCR. A Level Computer Science H446 Specification, Version 3.0, April 2026. Sections 3.2 to 3.4 and mark grids on pages 13 to 14 and 23 to 25. https://www.ocr.org.uk/Images/170844-specification-accredited-a-level-gce-computer-science-h446.pdf

OCR. Examiners' Report and Moderators' Report, H446/03/04, Summer 2025. Advice on contemporaneous development journals, testing evidence, failures, remediation and separation from final evaluation. https://www.ocr.org.uk/Images/752068-moderators-report.pdf

OCR. Moderators' Report, H446/03/04, Summer 2024. Advice on evidence throughout the lifecycle, robustness and usability testing. https://www.ocr.org.uk/Images/726389-moderators-report.pdf

OCR. Programming Project Set A High, 2017. Examiner commentary used only to identify evidence patterns; no candidate prose, code, screenshots or table content has been copied. https://www.ocr.org.uk/Images/514655-programming-project-set-a-high.pdf

OCR. Project Setting Guidance. Guidance on iterative testing, documenting failures/corrections and stakeholder involvement. https://www.ocr.org.uk/Images/324587-project-setting-guidance.pdf

CS Revise. OCR A Level Computer Science Component 03. Secondary checklist used as supporting guidance, not as an assessment authority. https://csrevise.co.uk/alevel/component03.html

Project-specific sources: PawMatch design draft 2 and the current Python/Streamlit/SQLite implementation, validation rules, schema, matching documentation and automated tests in the NEA repository.

Academic integrity note: public exemplar material was used to understand assessment structure and examiner expectations. The planned cases and wording in this document were created specifically for PawMatch.

Analysis Evidence Sources

The test plan is not based only on technical assumptions. The planned tests trace back to the three evidence types used in my analysis: competitor research, the client interview and the 27-response stakeholder survey. The codes below use the same E, Q and S prefixes as my analysis.

Competitor research source key

Code	Source	Finding carried into PawMatch	Test-plan areas supported
E1	Dogs Trust	Detailed profiles, reusable adopter information, organised animal cards, consistent headings and identifiable actions.	Navigation, animal cards/profiles, questionnaire journey and usability.
E2	RSPCA	Multiple species, clear categories, labelled filters, individual need profiles, consistent interface and staff-led final decisions.	Species/filter tests, profile retrieval, navigation and guidance wording.
E3	Petfinder	Step-by-step adopter questions, compatibility comparison, detailed profiles and clear matching actions.	Questionnaire progression, matching, result explanations and usability.

Client interview source key

Code	Interview evidence used	Test-plan connection
Q1	Adopters may overlook space, time, cost and experience.	Questionnaire validation and weighted compatibility cases.
Q2	Animal records need identity, photo, personality, medical, exercise and suitability details.	Admin record validation, profile selection and data-retrieval tests.
Q3	Warn and explain unsuitable matches, but keep staff as final decision maker.	Unmet-requirement explanations, cutoff and guidance tests.
Q5	Staff need to add, edit, remove and keep animal records current.	Administrator CRUD, status-change, persistence and deletion tests.
Q6	The interface should be simple, clear and quick for staff and adopters.	Navigation, clear-filter, state and final usability tests.
Q8	Users should search/filter animals, especially by type and size.	Browse search, lookup, combined-filter and empty-state tests.
Q9	Registration should collect core account details and age.	Registration normal, boundary and erroneous-data tests.
Q10	Welfare factors should carry more weight than softer preferences.	Individual weights, threshold, preference separation and ranking tests.
Q12	Passwords and account data need protection; only staff may alter animal records.	Hashing, authentication, direct-URL denial, role and user-isolation tests.
Q14	Results should be clear and lead to further animal information.	Ranked-result count/order, explanations and profile-link journey.
Q15	Compatibility scoring and accurate, current animal profiles are essential.	Database integrity, persistence, retrieval and end-to-end algorithm tests.

Stakeholder Survey Evidence

The survey had 27 responses. These codes justify why the plan tests different experience levels, detailed profiles, lifestyle factors, usability, scoring and welfare guidance rather than testing only whether pages load.

Stakeholder survey source key

Code	Survey evidence used	Test-plan connection
S1	Respondents included people with and without pet ownership experience.	Adopter usability should include inexperienced as well as experienced users.
S2	Few respondents had adopted from a centre.	Home/navigation tests must confirm the process and next actions are understandable.
S3	Lifestyle compatibility was the most common way respondents would choose a pet (48.1%).	Questionnaire and ranking tests focus on suitability, not appearance alone.
S4	96.3% rated research before adoption important or very important.	Animal profile completeness and accurate database retrieval are tested.
S5	Living space (88.9%) and time available (85.2%) were the most selected matching factors; garden access and practical care also ranked highly.	Boundary and algorithm-oracle tests cover housing, hours alone, activity and garden weighting.
S6	Responses placed high importance on ease of use.	Navigation, clear controls, state, browser/device and stakeholder usability tests.
S7	77.8% said a compatibility score would influence their decision.	Score calculation, category, ordering and explanation tests.
S8	70.4% supported adoption centres using compatibility scoring, but not everyone agreed.	Results must present scores as guidance and keep staff as the final decision maker.
S9	Responses showed concern for animal welfare.	Threshold, unsuitable-match, availability and integrity tests protect welfare-focused recommendations.

Traceability from analysis to planned tests

PawMatch requirement	Analysis sources	Planned test groups
R1 User and administrator authentication	Q9, Q12, Q15, S6	A01-A08, N01, N04, I04 and I07
R2 Homepage, navigation and clean interface	E1, E2, E3, Q1, Q6, Q7, Q11, Q13, S1, S2, S6	N01-N05, I01, I03-I04, P04-P05 and P07
R3 Administrator animal management	Q2, Q5, Q12, Q15	D01-D11, I02, I05 and I07
R4 Animal browsing and profiles	E1, E2, E3, Q2, Q6, Q8, Q14, Q15, S4, S6	B01-B08, I01-I03, P04 and P05
R5 Questionnaire and recommendations	E1, E3, Q1, Q3, Q10, Q14, Q15, S3, S5, S7, S8, S9	Q01-Q08, M01-M11, I01-I04, P01-P03, P05 and P08
Database integrity and persistence	Q5, Q10, Q12, Q15	D08-D11, Q06-Q07, M11 and I03-I07

Source limitation: the stakeholder survey contained 27 responses, so its findings guide the chosen tests but do not represent every adopter. Final usability evidence should therefore include the actual intended adopter and administrator stakeholders.